

TLEMCEN ZENATA, Algeria

WMO: 605310

Lat: 35.017N

Lon: 1.450W

Elev: 248

StdP: 98.38

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	3.6	-2.0	3.3	10.1	-0.2	3.8	10.2	12.2	14.7	10.6	14.0	0.7	120	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	36.5	21.3	34.7	21.1	33.0	21.1	24.5	31.4	23.8	30.3	23.1	29.3	4.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	27.4	22.0	17.2	26.5	21.2	16.3	26.0	75.5	31.5	72.4	30.2	69.7	29.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.9	6.9	DB	0.2	41.6	1.3	2.3	-0.8	43.2	-1.5	44.5	-2.3	45.8	-3.2	47.5	(4)
(5)			WB	-0.6	25.8	1.5	1.0	-1.7	26.5	-2.5	27.1	-3.4	27.6	-4.4	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	11.3	11.9	13.9	15.9	19.3	23.1	26.3	26.8	23.6	20.3	15.3	12.3	(6)
(7)		DBStd	5.98	2.49	2.59	2.45	2.31	2.62	2.43	2.24	2.10	2.19	2.68	3.01	2.64	(7)
(8)		HDD10.0	36	15	10	2	0	0	0	0	0	0	0	1	8	(8)
(9)		HDD18.3	935	219	181	138	79	19	0	0	0	0	11	101	188	(9)
(10)		CDD10.0	3091	54	63	123	177	288	394	506	522	407	319	159	79	(10)
(11)		CDD18.3	949	0	1	1	6	49	144	247	263	158	71	9	1	(11)
(12)		CDH23.3	9236	2	6	36	106	475	1374	2628	2790	1219	523	71	6	(12)
(13)	CDH26.7	4006	0	2	7	22	139	533	1312	1389	434	154	14	1	(13)	
(14)	Wind	WSAvg	2.2	2.4	2.4	2.2	2.3	2.4	2.4	2.2	2.0	1.9	2.3	2.1	(14)	
(15)	Precipitation	PrecAvg	442	47	50	55	50	41	10	2	3	15	34	48	61	(15)
(16)		PrecMax	1208	140	342	235	218	369	85	39	18	100	149	155	277	(16)
(17)		PrecMin	139	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	231	35	51	51	47	60	17	6	4	19	35	34	64	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	23.4	27.0	29.2	33.0	36.2	39.7	40.0	35.0	33.8	28.8	23.6	(19)
(20)			MCWB	13.4	13.8	16.2	17.0	18.7	20.5	22.1	22.1	21.3	18.2	17.5	14.9	(20)
(21)		2%	DB	20.0	21.0	23.8	26.1	30.1	33.5	36.6	36.6	32.2	30.2	25.2	21.0	(21)
(22)			MCWB	12.8	13.5	15.2	16.3	18.2	20.4	21.4	21.7	20.5	18.6	15.8	13.9	(22)
(23)		5%	DB	18.1	19.1	21.9	24.0	27.9	31.2	34.7	34.4	30.8	28.1	23.0	19.1	(23)
(24)			MCWB	12.2	12.7	14.6	15.6	17.7	20.2	21.1	21.6	20.6	18.6	15.3	13.1	(24)
(25)	10%	DB	16.8	17.4	19.9	22.1	26.0	29.8	32.8	32.9	29.1	26.1	21.1	17.7	(25)	
(26)		MCWB	11.7	12.3	14.0	15.1	17.2	19.8	21.0	21.8	20.5	18.3	14.9	12.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.8	18.1	19.1	21.2	23.5	25.4	25.9	24.0	21.7	19.3	16.5	(27)
(28)			MCD B	19.4	20.5	23.7	25.4	28.3	31.1	33.5	32.5	30.1	27.8	23.6	21.2	(28)
(29)		2%	WB	13.9	14.5	16.6	17.5	19.8	22.5	24.2	24.9	23.0	20.7	17.6	15.0	(29)
(30)			MCD B	17.9	18.8	21.9	23.3	26.6	30.0	31.8	31.6	28.7	26.4	22.6	18.9	(30)
(31)		5%	WB	13.0	13.5	15.4	16.6	18.9	21.6	23.5	24.2	22.3	19.9	16.6	13.9	(31)
(32)			MCD B	17.1	17.6	20.3	22.4	25.3	28.6	30.6	30.4	27.5	25.0	21.2	17.9	(32)
(33)	10%	WB	12.1	12.7	14.4	15.6	18.0	20.9	22.7	23.5	21.6	19.3	15.6	13.0	(33)	
(34)		MCD B	15.9	16.6	18.7	20.8	24.0	27.2	29.2	29.4	26.5	24.2	19.9	16.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	10.8	11.7	12.0	12.2	12.5	12.8	12.5	11.8	11.9	10.7	10.5	(35)
(36)			MCDBR	13.3	13.8	15.4	16.1	16.2	15.1	15.9	15.1	13.9	14.7	13.7	12.6	(36)
(37)			MCWBR	7.7	7.9	8.4	8.1	7.2	6.2	6.2	6.1	6.0	6.4	7.0	7.2	(37)
(38)		5% WB	MCD BR	11.3	11.6	13.1	13.3	13.1	13.1	12.7	12.1	11.4	12.1	11.5	10.9	(38)
(39)			MCWBR	7.2	7.3	8.0	7.3	6.4	5.6	5.3	5.3	5.5	5.9	6.5	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.343	0.405	0.430	0.467	0.500	0.553	0.524	0.474	0.414	0.360	0.325	(40)	
(41)		taud	2.507	2.426	2.225	2.169	2.090	2.002	1.871	1.952	2.092	2.255	2.410	2.499	(41)	
(42)		Ebn at Noon	892	899	863	857	827	797	751	765	785	805	825	855	(42)	
(43)		Edh at Noon	83	100	133	148	162	177	201	182	150	118	91	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.84	3.74	4.98	6.07	6.93	7.66	7.55	6.82	5.54	4.21	3.01	2.59	(44)	
(45)		RadStd	0.25	0.32	0.33	0.43	0.49	0.24	0.19	0.21	0.24	0.27	0.25	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page